

Research Paper

Strawberry disease detection using transfer learning of deep convolutional neural networks

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ABSTRACT

The impact of disease on strawberry quality and yield holds considerable significance, prompting researchers to explore effective methodologies for disease detection in strawberries. Among these, deep learning has emerged as a pivotal approach. In this regard, this research explored the utilization of transfer learning in deep convolutional neural networks (CNNs) to identify various strawberry diseases. Specifically, we utilized models pre-trained on the ImageNet dataset, namely VGG19, Inception V3, ResNet50, and DenseNet121 architectures, employing both fine-tuning and feature extraction techniques of transfer learning and consequently compared to the models without transfer learning. The target diseases for identification included angular leaf spot, anthracnose, gray mold, and powdery mildew on both fruit and leaves. The study outcomes revealed that Resnet-50 consistently achieved the highest accuracy across all three configurations, achieving its peak accuracy at 94.4 %, followed by Densenet-121 with an accuracy of 94.1 % attained through fine-tuning. These results highlighted the superior performance of fine-tuned models over using these models solely as feature extractors for identifying strawberry diseases. Furthermore, this study revealed that the application of transfer learning substantially reduced training time and resulted in a lower count of trainable parameters than models trained without transfer learning. These outcomes strongly endorse the practicality and effectiveness of employing transfer learning techniques for precise strawberry disease identification. Additionally, further research can explore the application of transfer learning to a broader range of crops and diseases, potentially enhancing agricultural disease detection methodologies.

1. Introduction

Strawberries rank among the most favored fruits and are extensively grown across the globe. Over the period of 2015 to 2021, annual worldwide production increased by approximately 11.6 %, with 9.17 million tons of production in 2021, as reported by FAOSTAT (2021). This increased production can be attributed to improved cultivation practices, especially within greenhouses. Greenhouses provide favorable environmental conditions, enabling strawberries to be cultivated with higher densities even in cold winter climates, which in turn significantly increases the yield and extends the supply period (Hernández-Martínez et al., 2023; Laugale et al., 2017). However, the controlled environmental parameters such as temperature and humidity inside

greenhouses for extended periods can promote the proliferation of pests and diseases. Furthermore, the prolonged production cycle and continuous fruiting not only increase the potential incubation period of pathogens within the environment but also create a scenario where infection from one pathogen may make plants vulnerable to subsequent infections by other pathogens, thereby increasing the risks (Lu et al., 2017; Zhao et al., 2022). If not promptly identified and controlled, diseases have the potential to disrupt the strawberry production cycle, leading to inconsistent yields. Furthermore, they can diminish both the quantity and quality of the fruits and their post-harvest lifespan (Hernández-Martínez et al., 2023). Therefore, the timely identification and prevention of disease incidence and development within the production environment becomes necessary.

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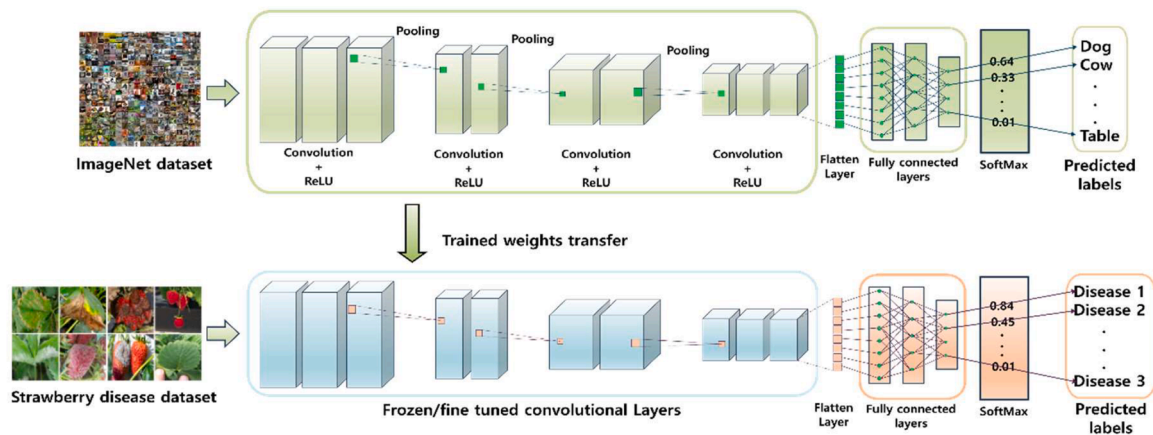


Fig. 1. Schematic diagram of the transfer learning process of deep learning architectures.

Strawberries affected by diseases exhibit unique visible marks and lesions on either the leaves, stems, flowers, and/or fruits, which can be used to diagnose the diseases (Ngugi et al., 2021). Traditionally, disease diagnosis by extension officers relied on visual inspection or laboratory tests, methods demanding considerable expertise, and often involving high costs and time consumption (Liu et al., 2017; Wang et al., 2012). Moreover, trained experts might make incorrect diagnosis when faced with diseases that share visually similar characteristics, and individual experts often specialize in a limited range of diseases (Barbedo, 2018; Cruz et al., 2017). Loop-mediated isothermal amplification (LAMP), enzyme-linked immunosorbent assay (ELISA), and polymerase chain reaction (PCR) are some particular laboratory-based molecular techniques commonly employed for detecting strawberry diseases (Beckman et al., 1994; Mirmajlessi et al., 2015; Ren et al., 2021; Zhang et al., 2016; Zimmermann et al., 2004). Nevertheless, these methods have limitations, such as being time-consuming, labor-intensive, and requiring complex procedures, particularly during sample preparation, to yield reliable and accurate outcomes. Moreover, these techniques necessitate specific reagents tailored for each targeted pathogen. While molecular techniques are robust for confirming the presence of plant diseases, they are not ideal as primary screening tool for handling a substantial volume of plant samples due to the resource-intensive nature of the associated procedures (Sankaran et al., 2010). Various spectroscopic and imaging techniques such as multispectral or hyperspectral imaging and infrared spectroscopy for large-scale disease detection under field conditions have been developed in recent years which are rapid, accurate, and can be performed remotely; however, these are still expensive and disease-specific (Sankaran et al., 2010). Thus, an essential requirement is a rapid, precise and accurate strawberry disease detection system that can identify multiple diseases with similar visual characteristics.

In recent years, progress in image detection has been significantly accelerated by the advancements in computer vision technologies. The utilization of machine learning or deep learning is progressively becoming a standard solution in plant disease detection systems (Zhao et al., 2022). Machine learning approaches employ hand-crafted features extracted from techniques like gray level co-occurrence matrix (GLCM), histogram-oriented gradients (HOG), scale invariant feature transform (SIFT), and others to identify plant diseases. However, the challenge lies in the difficult and time-consuming nature of feature extraction in real field environments, rendering machine learning less adaptable for crop disease detection, as highlighted by Shin et al. (2021) and Zhao et al. (2022). On the contrary, deep learning, particularly convolutional neural networks (CNNs), can identify and extract features automatically during the learning process, making it suitable for tasks like plant disease detection. Ramcharan et al. (2017) utilized the InceptionV3 CNN model to accurately identify three disease classes, two mite damage classes, and a healthy class in cassava plants, achieving an

overall accuracy of 93 %. Similarly, Ferentinos (2018) achieved an overall accuracy of 99.53% in classifying 58 different classes of (plant, disease) combinations of 25 plant species. Likewise, AlexNet and GoogleNet architectures were used to identify 26 different diseases from 14 crop species, achieving an accuracy of 99.35 % by Mohanty et al. (2016). Although very good results have been achieved, significant volume of training data is required for CNNs to effectively learn underlying features at varying levels and achieve a high-performance model (Karimi et al., 2019; Li et al., 2018). Ferentinos (2018), Mohanty et al. (2016), and Khan et al. (2022) used 87,848, 54,306, and 9000 images, respectively, to train their deep learning models. In the context of plant disease detection, it is challenging when it comes to gathering such a substantial amount of correctly labeled data. This difficulty arises from factors such as the time-consuming nature of collecting labeled datasets, which necessitates the involvement of experts to identify and label the diseases accurately. Additionally, obtaining a substantial number of samples infected with a single disease becomes challenging. Despite the limitations, the prior studies have effectively showcased the capability of deep learning algorithms in detecting plant diseases.

An approach to overcome the drawbacks of requiring a large dataset to train a CNN model is to employ transfer learning. Transfer learning refers to applying knowledge acquired from one task to another similar task (Alzubaidi et al., 2021). The diagrammatic representation of the transfer learning process is shown in Fig. 1. With transfer learning, model parameters can either be initialized from those of pre-trained networks and subsequently fine-tuned on new tasks or freeze those parameter weights, utilizing them as a feature extractor (Chen et al., 2020; Coulibaly et al., 2019). In this study, our proposed approach for developing a strawberry disease detection model using deep learning with transfer learning involves two methods: feature extraction and fine-tuning. Furthermore, we undertake training without transfer learning and perform a comparative analysis of the model performances with and without transfer learning. Consequently, the objectives of this study encompass: (1) developing an optimized deep learning model with transfer learning to identify six strawberry diseases and two healthy classes; (2) evaluating the performance and effectiveness of pre-trained deep learning models developed using transfer learning techniques, in addition to models without transfer learning; (3) comparison of different model performance in terms of classification metrics and, computational time.

2. Materials and methods

2.1. Image datasets

During this study, a dataset was created comprising images of strawberry fruits and leaves affected by six distinct diseases alongside

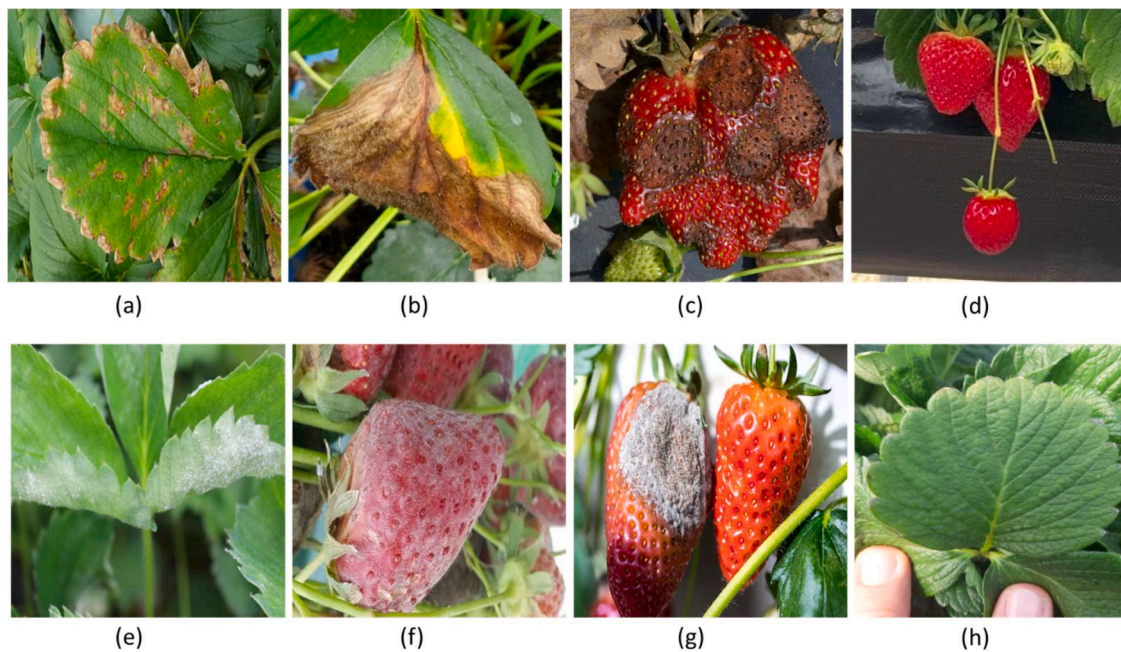


Fig. 2. Healthy and diseased strawberry leaves and fruits. (a) Angular leaf spot, (b) Gray mold on leaf, (c) Anthracnose, (d) Healthy Fruit, (e) Powdery mildew on leaf, (f) Powdery mildew on fruit, (g) Gray mold on fruit, (h) Healthy leaf.

images of healthy fruits and leaves collected from two different sources: field collection and publicly available online resources. The diseases represented in the dataset include angular leaf spot, anthracnose, gray mold on a leaf, gray mold on fruit, powdery mildew on a leaf, and powdery mildew on fruit, along with the healthy fruit and leaf samples, as illustrated in Fig. 2. Specifically, images featuring healthy plants and gray mold disease were acquired from the experimental greenhouse of the Smart Space Sensing Laboratory at Gyeongsang National University in South Korea. To accomplish this, strawberry plants were inoculated with a fungus pathogen (*Botrytis cinerea*), and images were captured once visual symptoms appeared on leaves and fruits. Furthermore, images with anthracnose and angular leaf spots are sourced from the strawberry farms around Jinju City, Gyeongsangnam-do, South Korea. For powdery mildew images, an open-source dataset shared by Afzaal et al. (2021) from Jeonbuk National University, South Korea, was utilized. To replicate field conditions, all images were captured under non-uniform illumination intensities and diverse background conditions. Finally, to ensure the accuracy of annotations, the images were annotated through a collaborative effort involving farmers and were subsequently subjected to validation by the plant pathologist from the Department of Plant Pathology at Gyeongsang National University. During the validation process, samples of diseased fruit and leaves were collected along with the image acquisition and taken to the laboratory. Initially, diseases were identified based on visual symptoms, which were subsequently confirmed through corresponding laboratory tests.

Data augmentation is a method employed to artificially expand the size of a training dataset by implementing various transformations or modifications to the existing data (Maharana et al., 2022). The size of the available dataset influences the performance of CNN models, and increasing the dataset size can lead to improvements in model performance (Khosla and Saini, 2020; Maharana et al., 2022). Moreover, Khosla & Saini (2020) reported that data augmentation can effectively mitigate overfitting, further underscoring its significance. Hence, data augmentation was applied to the original dataset to increase the number of observations and improve the architectures' robustness. To achieve this, several transformations were applied to the original images. Images were rotated by 30, 60, 90, 120 and 150° clockwise. Random zooming was also applied, with a zoom range set to 0.2. Moreover, Gaussian blur was applied with a kernel size of 3×3 and sigma values ranging

between 0.1 and 2 to introduce some noise into the images. By employing these transformations, the original dataset (D1) of 3271 images was augmented to produce 22,897 images. Finally, as per the objective to compare the performance of models with and without transfer learning, another dataset referred as augmented dataset (D2) was created combining original and augmented images.

The datasets' images were pre-processed using Irfanview image processing software. Initially, all images were cropped to 420×420 pixels and resized to 224×244 pixels, or 299×299 pixels, depending on specific architecture requirements. VGG19, ResNet50, and DenseNet121 necessitated an input size of 224×224 pixels, whereas InceptionV3 required an input size of 299×299 pixels. Moreover, a crucial aspect of the pre-processing involved normalization. This normalization process was carried out utilizing the PyTorch deep learning framework, aligning the image data with the mean and standard deviation values of the ImageNet Dataset. Specifically, the mean values for the Red (R), Green (G), and Blue (B) color channels were set to 0.485, 0.456, and 0.406, respectively, while the standard deviation values for these channels were set to 0.229, 0.224, and 0.225 (Nguyen et al., 2019; Russakovsky et al., 2015).

2.2. Overview of pre-trained CNN architectures

Convolutional neural network (CNN) is a deep learning architecture designed to work with grid-structured data with spatial relationships in the grid's local regions (Shanthamallu and Spanias, 2022). CNNs utilize convolutional layers that apply filters or kernels to small local regions of the input grid, enabling the detection of specific features or patterns within these regions (Alzubaidi et al., 2021). Throughout this process, the network captures spatially related information and learns hierarchical representations of the input data. This ability to exploit local spatial dependencies makes CNNs highly effective in image recognition, object detection, and image segmentation tasks.

ImageNet Large Scale Visual Recognition Challenge (ILSVRC) is one of the image recognition competitions where various CNN architectures have achieved a classification error rate of less than 4% in the ImageNet dataset. Such architectures have been successfully employed for different recognition tasks with improved generalization and convergence. Among them, Alexnet, Squeezenet, Resnet, Visual geometry

group (VGG), Googlenet, Inception:Resnet and Inception V3/4, DenseNet, and Xception are some architectures that have been employed in various domains, showcasing their efficacy in tasks such as detecting millet crop diseases (Coulibaly et al., 2019), different plant diseases (Chen et al., 2020; Too et al., 2019), coconut pest and diseases (Singh et al., 2021) and powdery mildew disease in strawberry leaves (Shin et al., 2021). Therefore, this study selected four architectures: ResNet50, VGG19, Inception V3 and DenseNet121 for strawberry disease detection. Each of these architectures has been widely utilized in plant disease detection and each architecture is different in design characteristics. In the subsequent section, we will discuss some of the architectures employed in this study in detail.

2.2.1. ResNet50

Resnet-50 (He et al., 2016) is a convolutional neural network, featuring 48 convolutional layers, one max-pooling layer, and one average pooling layer. The convolutional layers are grouped into four stages, each with a varying number of residual blocks (3,4,6 and 3). Each residual block comprises three convolutional layers with kernel sizes 1×1 , 3×3 , and 1×1 , respectively. The novelty of Resnet is the incorporation of residual blocks, which mitigates the vanishing gradient problems in deep neural networks. In a residual block, the original input to the block is added (shortcut connection) to the output of the last 1×1 convolution. This addition operation allows the model to skip over some layers and propagate the gradient more effectively (Mascarenhas and Agarwal, 2021).

Resnet50 being deep architecture, utilizes depth to learn complex patterns and features in input images while minimizing vanishing gradient problem efficiently through residual connections. The use of residual connections also facilitates faster convergence and performance. However, Resnet50 requires more computation and poses a risk of overfitting due to its large number of parameters.

2.2.2. VGG19

VGG19 (Simonyan and Zisserman, 2014) is composed of 19 layers, including 16 convolutional layers and 3 fully connected layers. The architecture follows a sequential structure, with convolutional layers stacked on top of each other, followed by three fully connected layers for classification. VGG19 utilizes 3×3 convolutional filters throughout the network, which is a small receptive field size, enabling the network to capture intricate details in the input images. Furthermore, the stacking of convolutional layers enhances the network's depth, enabling it to learn more complex features. To down-sample feature maps, max pooling is applied after every two convolutional layers. The final layer of the model is a Softmax layer, which produces classification probabilities. A distinctive feature of the VGG19 architecture is its parameter sharing strategy, where the same filter weights are shared across all spatial locations within a layer (Vignesh et al., 2023).

VGG19 presents a straightforward architecture consisting of repeated convolutional and pooling layers, making it easy to understand and implement. Despite its simplicity, VGG19 has demonstrated competitive performance in image recognition tasks like ILSVRC. Nevertheless, VGG19 has a relatively large number of parameters, resulting in higher memory footprint and computational requirements compared to more compact models.

2.2.3. Inception V3

Inception V3 proposed by Szegedy et al. (2016) is an extension of the inception architecture series designed for image classification and recognition tasks. The Inception V3 architecture follows a pattern of factorizing larger convolutions into smaller ones, reducing the number of parameters and computational costs (Singh et al., 2021). This architecture consists of inception modules composed of convolutional layers arranged in parallel. These layers have different filter sizes (1×1 , 3×3 , and 5×5) along with pooling operations. The outputs of these parallel operations are then combined, allowing the network to capture

information at different scales and with different receptive field sizes. Moreover, Inception V3 incorporates auxiliary classifiers at intermediate layers, strategically placed to address the vanishing gradient problem during training, allowing improved gradient flow, regularization, and overall training performance (Szegedy et al., 2016).

Inception V3 captures features at multiple scales using inception modules, enabling the model to effectively handle objects of different sizes and orientations while maintaining high computational efficiency. Nonetheless, the design of Inception modules and the incorporation of various kernel sizes increase the complexity of the model, thereby making it challenging to interpret and implement compared to simpler architectures.

2.2.4. DenseNet121

DenseNet121, introduced by Huang et al. (2017), is a specific variant of the DenseNet architecture, named after its depth of 121 layers. The architecture comprised a sequence of dense blocks followed by transition layers, each containing multiple layers. Specifically, DenseNet121 is configured with four dense blocks, and each dense block contains multiple layers. Typically, a single layer within a dense block consists of a batch normalization layer, a convolutional layer, and an activation function. The key characteristic of dense blocks is the dense connections, where each layer is connected to every other layer within the same block. This dense connectivity fosters collective knowledge utilization from all preceding layers, promoting feature reuse and effective information flow (Alzubaidi et al., 2021). Transition layers are inserted between dense blocks to facilitate dimensionality reduction, reducing the number of feature maps and spatial dimensions. Each transition layer typically includes a batch normalization layer, a 1×1 convolutional layer, and an average pooling operation. The purpose of the transition layers is to control the number of parameters and computational complexity while maintaining information flow (Vellaichamy et al., 2021).

DenseNet121 has high parameter efficiency achieved by encouraging feature reuse and reducing redundancy, resulting in models with fewer parameters compared to traditional architectures. Additionally, DenseNet121 effectively mitigates the risk of overfitting and improves generalization performance through dense connectivity. However, the dense connectivity pattern requires storing feature maps from all preceding layers, leading to higher memory consumption during training and inference.

2.3. Transfer learning

Transfer learning allows the application of knowledge acquired during the training of a model for a particular task to be carried over and applied to another related task (Alzubaidi et al., 2021). Instead of starting the model training from scratch, transfer learning permits the utilization of pre-trained models and their learned features. Notable examples of such models include Resnet50, DenseNet121, inceptionV3, and VGG19, which have been pre-trained on the ImageNet dataset for image classification. ImageNet is a large-scale, publicly available dataset comprising approximately 15 million images across 22,000 categories, specifically designed for visual object recognition. In this study, we employed pre-trained models with transfer learning using two common techniques: feature extraction and fine-tuning. The subsequent section explains the process in detail.

2.3.1. Feature extraction

The feature extraction method is a common approach employed in transfer learning wherein a pre-trained deep learning model serves as a feature extractor. In this approach, convolutional layers are frozen with the weights obtained from training on large datasets like ImageNet dataset. By freezing the feature extractor layers, models can leverage pre-trained weights and capture relevant features specific to the strawberry disease domain. Subsequently, the classifier layers are trained on

features extracted from a new dataset using frozen layers. In this study, we imported pre-trained deep learning models discussed in section 2.2 from the PyTorch framework 2.1.0, all initially trained on the ImageNet dataset. The convolutional base of the network was set as non-trainable, ensuring the pre-trained weights were retained and not updated during the training process. On top of the frozen convolutional base, the output of the classifier layer was modified to give the output equal to the number of classes specific to this study rather than the original 1000 classes of ImageNet. The models developed to use as feature extractors are designated as Type 1 models within this study.

2.3.2. Fine-tuning

Fine-tuning is another commonly employed technique in transfer learning. Unlike the feature extraction technique, which keeps the convolutional base frozen, fine-tuning involves updating the weights of certain or all layers of the pre-trained model to customize it to a new task or dataset. In this study, similar to the feature extraction technique, models mentioned in section 2.2 were imported from the PyTorch framework 2.1.0, and the output layer of the classifier part was modified to produce the number of output classes specific to this study. Subsequently, the entire model underwent training, starting from the weights of the convolutional layers initialized with the pre-trained weights. This approach facilitated the acquisition of task-specific features while leveraging the existing knowledge embedded in the convolutional base. These fine-tuned models are categorized as Type 2 models within this study.

2.4. Experimental setup

All models detailed in section 2.2 were imported from the PyTorch deep learning framework and were configured with a cross-entropy loss function. To determine the optimal optimizer and learning rate for the developed models, grad student descent (GSD) was employed, following the methodology outlined by Yang and Shami (2020). Subsequently, Stochastic gradient descent (SGD) was used as an optimizer with a learning rate set to 0.01. Each model underwent training for a total of 200 epochs, with the dataset split into 80 % for training, 10 % for validation, and 10 % for testing. The training and testing were conducted on hardware with a 2.90 GHz Intel(R) Core™ i7–10700 CPU, 10.0 GB RAM, and NVIDIA GeForce RTX 2060 with 8 GB of memory.

In order to further evaluate the performance of deep learning models with transfer learning, all four models were trained without transfer learning using another dataset having original and augmented images. Similar to Type 1 and Type 3 models, the last layer was modified to have an output equal to the eight classes rather than the original 1000 classes. The models without transfer learning were classified as Type 3 models.

2.5. Evaluation metrics

In this study, four distinct deep learning architectures, both with and without transfer learning, were used to identify eight classes of healthy and diseased strawberries and four performance metrics, namely, precision, recall, F1-score, accuracy, and area under the receiver operating characteristic curve (AUC ROC) were used to assess the performance of each model. Each metric is defined in equations (1) – (5), as explained by Y. Liu et al. (2014).

$$\text{Accuracy} = \frac{TP + TN}{M} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

Table 1

Summary of image datasets used for strawberry disease detection.

Disease Class	Original dataset (D1)	Augmented dataset (D2)
Angular Leafspot	420	3575
Anthracnose Fruit	417	3318
Gray mold Fruit	450	3565
Gray mold Leaf	374	3717
Healthy Fruit	370	2955
Healthy Leaf	395	2286
Powdery Mildew Fruit	330	2665
Powdery Mildew Leaf	515	4087

$$F1 - \text{Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

$$\text{Area under the ROC curve (AUC)} = \int_0^1 t \times df \quad (5)$$

Where M = the total number of samples, TP = true positives, TN = true negatives, FP = false positives, FN = false negatives, t = true positive rate, and f = false positive rate.

3. Results

3.1. Deep learning models with feature extraction

Table 1 summarizes the performance metrics of four different deep learning architectures whose convolutional layers were frozen with pre-trained weights from the ImageNet dataset, and the neural network was trained with our dataset. Among these architectures, Resnet-50 achieved the highest accuracy of 94.1 %, with precision and recall at 94.3 % and 94.1 %, respectively. Inception V3 and Densenet-121 exhibited equal accuracy of 91.5 %, while VGG19 lagged slightly behind with 90.6 % accuracy. This trend was consistent across precision, recall, and F1-score metrics.

To assess the diagnostic capability of each classifier, the receiver operating characteristic (ROC) curves were generated and calculated the area under the ROC curve (AUC). Table 1 contains the average AUC for each classifier, accompanied by other metrics, all exceeding 0.99, signifying a high ability to identify strawberry diseases correctly. Fig. 3 (a) shows the ROC curve for Resnet-50, which performed better compared to other architectures in terms of accuracy and F1-score. The class-wise classification results for Resnet-50 are tabulated in Table 2, and the confusion matrix in Fig. 4(a) shows the numbers of correct and incorrect classifications made by the model. For Resnet-50, a recall of 1 was achieved in identifying gray mold disease on leaves, indicating its high identification capability for this disease. However, a recall of only 0.882 was achieved for powdery mildew in fruit, making it the most challenging disease for the model to identify. The learning curves of all four Type 1 models are illustrated in Fig. 5.

Even though VGG19 comprised only 19 layers, it possessed the highest number of trainable parameters among all models, totaling 32.7×10^{-3} million. In contrast, Densenet121, with the highest number of layers, had the lowest count of trainable parameters at only 8.2×10^{-3} million. Resnet50 exhibited the shortest training time, requiring only 61.61 min, whereas InceptionV3 consumed the most at 85.53 min. Notably, both Resnet50 and InceptionV3 had an identical number of trainable parameters, underscoring the influence of model architecture on the training duration in deep learning models. A comprehensive breakdown of the depth, size, trainable parameters, and training time for various models is presented in Table 3.

3.2. Deep learning models with fine-tuning

All four deep learning models exhibited competitive performance in

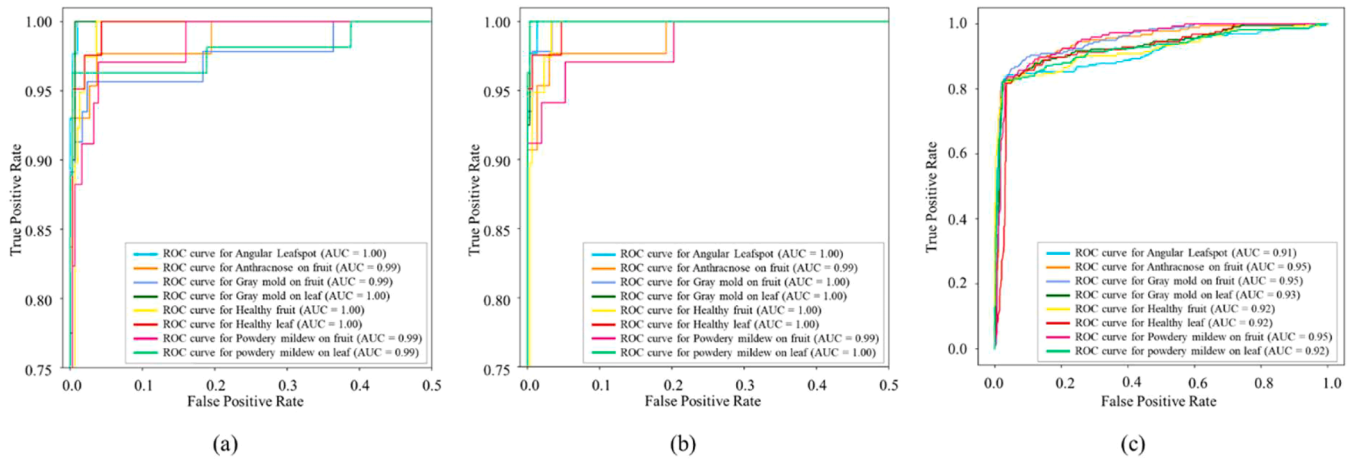


Fig. 3. Class-specific ROC curves for Resnet-50. (a) Feature extraction, (b) Fine tuning, (c) Without transfer learning.

Table 2

Performance metrics of four deep learning architectures with feature extraction.

Models	Accuracy	Precision	Recall	F1-score	AUC
Densenet-121	0.915	0.922	0.915	0.915	0.991
Inception V3	0.915	0.916	0.915	0.914	0.992
Resnet-50	0.941	0.943	0.941	0.941	0.993
VGG19	0.906	0.907	0.906	0.906	0.993

identifying strawberry diseases. Among the models, Resnet-50 achieved the highest accuracy of 94.4 %, a precision of 95.2 %, and a recall of 94.4 % in strawberry disease identification. Densenet-121 exhibited a similar performance to Resnet-50 with only a 0.3 % difference in accuracy. Similarly, inception V3 and VGG19 also demonstrated comparable results. However, inception V3 had the lowest performance, achieving an accuracy of 93.5 %.

The models Densenet-121, inception V3, and VGG-19 consistently performed well across accuracy, precision, recall, and F1-score metrics. In contrast, Resnet-50 stood out with the highest precision of 95.2 %, highlighting its ability to make accurate positive predictions. For a more comprehensive performance evaluation, ROC for all models was created, and the area under the curve (AUC) was calculated. Notably, all four models achieved an AUC score exceeding 0.99, indicating their ability to identify strawberry diseases accurately. Detailed performance metrics for each model are presented in Table 4.

Given that Resnet-50 was the top performing model in terms of accuracy and F1-score, class-wise performance was further analyzed.

Table 5 provides an overview of the class-specific performance metrics for Resnet-50. During the identification of powdery mildew in both fruit and leaves and anthracnose in fruit, the precision was 1.0, indicating that all predicted instances were true positives. In contrast, when identifying gray mold in fruit, the precision was only 0.78, implying that 78 % of instances predicted as gray mold in fruit were confirmed as actual instances of this disease. Regarding recall, gray mold disease in both fruit and leaves achieved a perfect recall of 1.0, showcasing the model's ability to accurately identify all instances of gray mold diseases. On the other hand, when identifying powdery mildew in fruit, the recall score was 0.794, indicating that the model correctly identified only 79.4 % of the total true positives for this disease. In classification, a trade-off occurs between the precision and recall as the model adjusts the classification threshold to optimize either metric. Increasing precision by setting high threshold in making positive predictions often results in more false negatives, reducing the recall. Conversely, lowering the decision threshold to increase recall tends to increase false positives, reducing precision. In this study, significant trade-offs between precision and recall, particularly in diseases powdery mildew and gray mold on fruit was observed.

The ROC curve and the confusion matrix for model Resnet-50 are depicted in Fig.3(b) and Fig. 4(b), respectively. The analysis of the confusion matrix unveiled misclassifications primarily occurring within two categories: leaf-related diseases and fruit-related diseases. This observation suggests that Resnet-50 possesses the ability to differentiate between diseases affecting leaves and those affecting fruits. The misclassifications are primarily due to the similarity in visual features

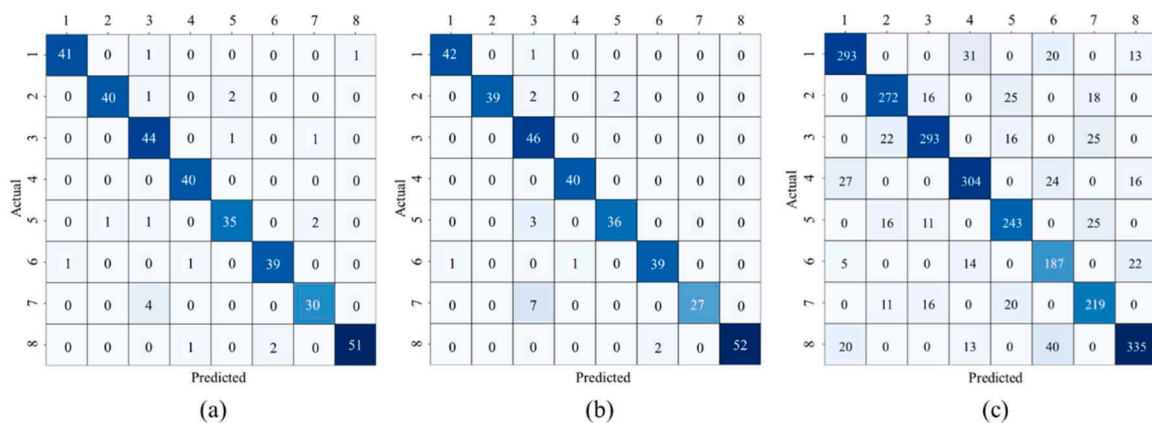


Fig. 4. Confusion matrix for Resnet-50 with (a) feature extraction, (b) fine tuning, (c) without transfer learning with classes (1) Angular Leafspot, (2) Anthracnose, (3) Gray mold on fruit, (4) Gray mold on leaf, (5) Healthy fruit, (6) Healthy leaf, (7) Powdery mildew on fruit, and (8) Powdery mildew on leaf.

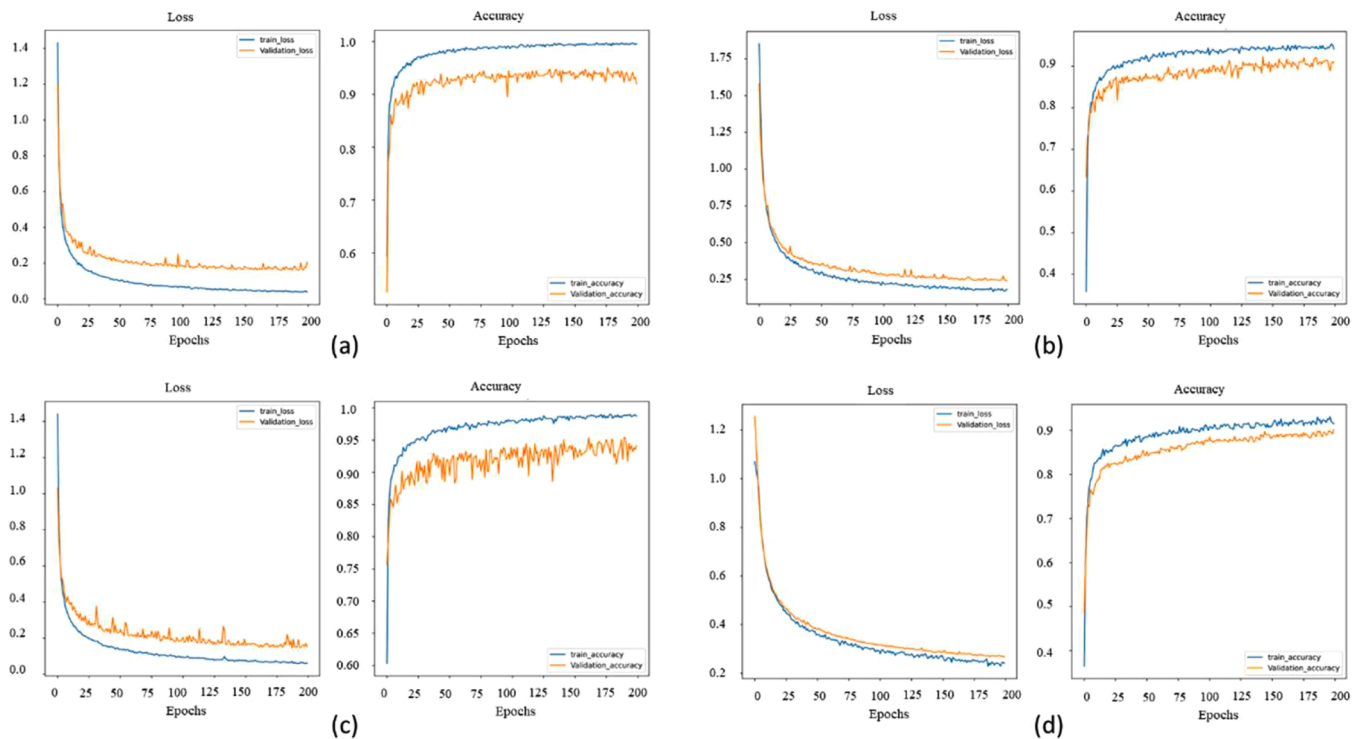


Fig. 5. Performance of four deep learning models with feature extraction. (a) ResNet-50, (b) Densenet-121, (c) VGG-19, (d) Inception V3.

Table 3

Class-specific performance metrics of Resnet-50 with feature extraction.

Classes	Precision	Recall	F1-score
Angular Leafspot	1.0	0.953	0.976
Anthracoese Fruit	0.868	0.767	0.815
Gray Mold Fruit	0.843	0.935	0.887
Gray Mold Leaf	0.93	1.0	0.964
Healthy Fruit	0.897	0.897	0.897
Healthy Leaf	0.975	0.951	0.963
Powdery Mildew Fruit	0.8	0.824	0.812
Powdery Mildew Leaf	0.981	0.963	0.972

Table 4

Performance metrics of deep learning architectures with fine-tuning.

Models	Accuracy	Precision	Recall	F1-score	AUC
Densenet-121	0.941	0.944	0.941	0.941	0.997
Inception V3	0.935	0.936	0.935	0.935	0.995
Resnet-50	0.944	0.952	0.944	0.945	0.998
VGG19	0.938	0.938	0.938	0.937	0.997

Table 5

Class-specific performance metrics of Resnet-50 with fine-tuning.

Classes	Precision	Recall	F1-score
Angular Leafspot	0.977	0.977	0.977
Anthracoese Fruit	1.0	0.907	0.951
Gray Mold Fruit	0.78	1.0	0.876
Gray Mold Leaf	0.976	1.0	0.988
Healthy Fruit	0.947	0.923	0.935
Healthy Leaf	0.951	0.951	0.951
Powdery Mildew Fruit	1.0	0.794	0.885
Powdery Mildew Leaf	1.0	0.963	0.981

among classes of leaves and fruits. Since deep learning models heavily rely on patterns and features extracted from input data to make predictions, the discriminative capacity for the model becomes limited when dealing with visually similar classes, thereby posing challenges in differentiation. The learning curves of Type 2 models are illustrated in Fig. 6.

In terms of model training time, VGG19 required the highest time at 161.69 min, primarily due to its highest count of trainable parameters with 139.6 million parameters. The best performing model, Resnet50, took only 105.93 min for training. The details regarding the depth, size, trainable parameters, and training time for Type 2 models are tabulated in Table 3.

3.3. Deep learning models with augmented images

In order to further evaluate the models with transfer learning, the models were trained from scratch with augmented images. The outcomes presented in Table 6 summarize the performance metrics for four different deep learning models trained without transfer learning. Among these models, Resnet-50 achieved the highest accuracy of 82.1 % and precision of 82.5 %. These figures signify not only its accuracy in making correct predictions but also its capability to minimize the occurrence of false positives. Moreover, Resnet-50 displayed a recall rate of 82.1 %, indicating its effectiveness in correctly identifying a significant portion of positive instances. In contrast, Inception V3 exhibited the lowest performance with an accuracy of 79.6 % and consistent values across the precision, recall, and F1-score metrics.

Similar to the models utilizing transfer learning, ROC curves for each model were created for further evaluation, calculating the corresponding AUC scores. In terms of AUC score, Resnet-50 demonstrated an average AUC score of 0.932, while the other three models exhibited scores ranging from 0.922 to 0.926, indicating Resnet-50's ability to distinguish positive and negative instances compared to other models. Fig. 3(c) graphically illustrates the class-specific ROC curves, and Fig. 4 (c) depicts the confusion matrix, showing the number of predictions made for each class by the best performing model, Resnet-50, in terms of

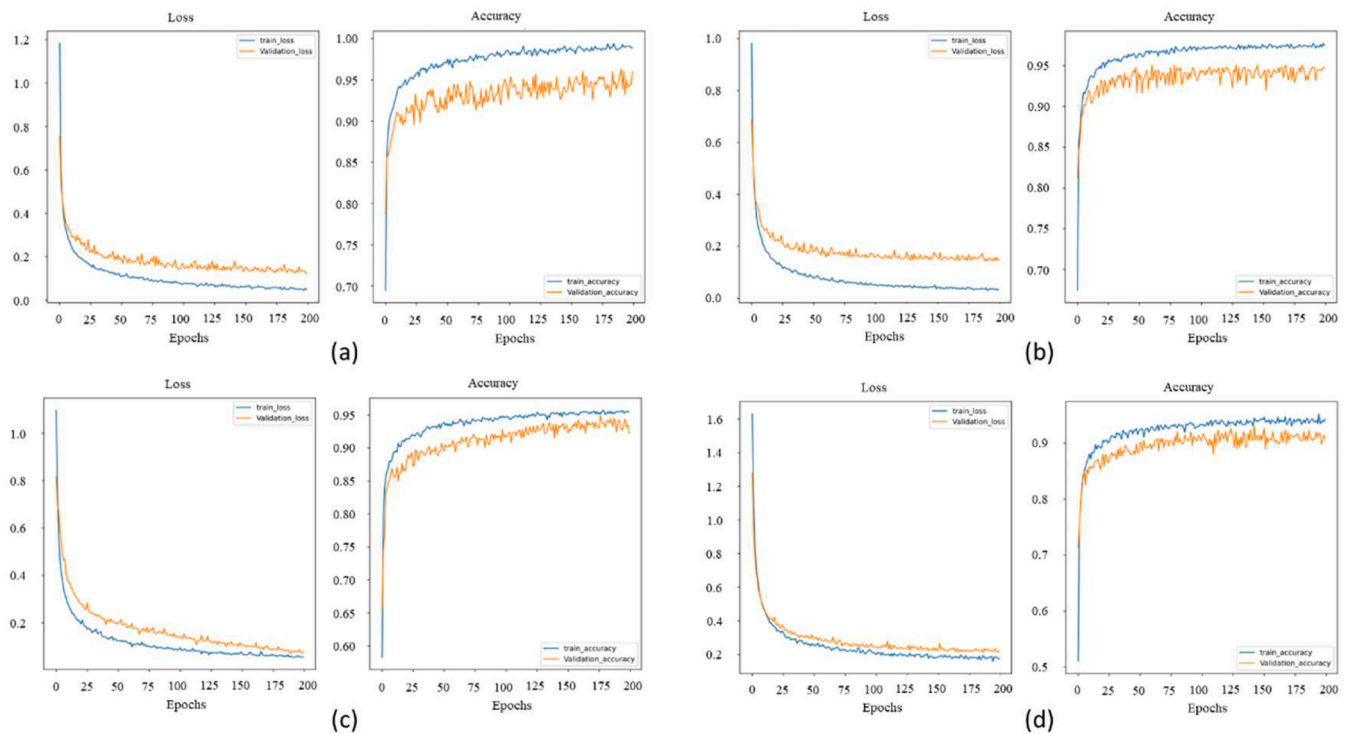


Fig. 6. Performance of four deep learning models with fine-tuning. (a) ResNet-50, (b) Densenet-121, (c) VGG-19, (d) Inception V3.

Table 6

Performance metrics of deep learning architectures without transfer learning.

Models	Accuracy	Precision	Recall	F1-score	AUC
Densenet-121	0.805	0.808	0.805	0.806	0.924
Inception V3	0.796	0.80	0.796	0.797	0.922
Resnet-50	0.821	0.825	0.821	0.822	0.932
VGG19	0.814	0.817	0.814	0.815	0.926

accuracy and F1-score metrics.

Gray mold on fruit was the most precisely detected disease class with a precision of 0.871, indicating 87.1 % of the instances predicted are true positives. In contrast, the precision for the healthy leaf class was only 0.691, suggesting that Resnet-50 had a 30.9 % likelihood of misclassifying other disease classes as healthy leaves. Regarding recall, values ranged from 0.817 to 0.823 across different classes, with the lowest for the healthy leaf and the highest for powdery mildew on fruit. This consistency implies that the model successfully identified diseases with more than 80 % of instances correctly identified. Moreover, all disease classes achieved an AUC score exceeding 0.90, indicating the model's strong discriminative capability. The learning curves for all Type 3 models are illustrated in Fig. 7.

The number of trainable parameters of Type 3 models was identical to those in Type 2 models. However, the training time of all models increased compared to Type 2 models due to the utilization of a larger dataset for training. InceptionV3 and Resnet50 required the highest and lowest training time, with 1622.03 and 1168.4 min, respectively. The specifics regarding the depth, trainable parameters, and training time for all Type 3 models are presented in Table 3.

4. Discussion

The timely and precise identification of diseases in strawberries cultivated under controlled environments has been crucial for upholding production quality and minimizing potential losses. Recent advances in deep learning and image processing have significantly improved their

applicability in strawberry disease detections. However, two critical factors, model architecture, and the dataset, significantly influence the performance of deep learning models. In domains like disease detection, designing an optimal architecture and training it with a large dataset proves to be a difficult task. In such scenarios, transfer learning, which has been validated as effective by various studies, becomes a valuable approach as it effectively addresses both challenges. Therefore, this study assessed the efficacy of four pre-trained deep learning architectures, adapted using transfer learning techniques, specifically fine-tuning and feature extraction. Subsequently, a comparative analysis was conducted to assess the performance of these models with transfer learning against models that were trained and tested using augmented images without transfer learning.

The results showed that the models with transfer learning performed better than those without. Notably, models with transfer learning had significantly reduced training times compared to their non-transfer learning counterparts. Resnet50 showed the highest accuracy and shortest training time in all three configurations, which may be attributed to its architecture, which utilizes skip connections to allow efficient gradient flow during backpropagation, preventing vanishing gradient problem (Alzubaidi et al., 2021). In contrast to Densenet-121, where skip connections link each layer to every other layer in the network, Resnet50's skip connections transmit input to the transformed output of residual blocks. This design enables the network to learn residuals rather than the complete transformation, preserving the input information (Bodhwani et al., 2019). Additionally, the bottleneck structure of the residual block in Resnet50 reduces the input dimensionality, resulting in fewer parameters and computations. Consequently, this reduction contributes to a decrease in the network's training time, as evidenced in our results.

The models with Type 1 configurations took the least amount of time, ranging from 61.61 min for Resnet-50 to 85.53 min for InceptionV3 to train, followed by the Type 2 configurations with a minimum training time of 105.93 min for Resnet50 even though the same dataset was used for training. This contrast in training times can be attributed to significant differences in the number of trainable parameters between

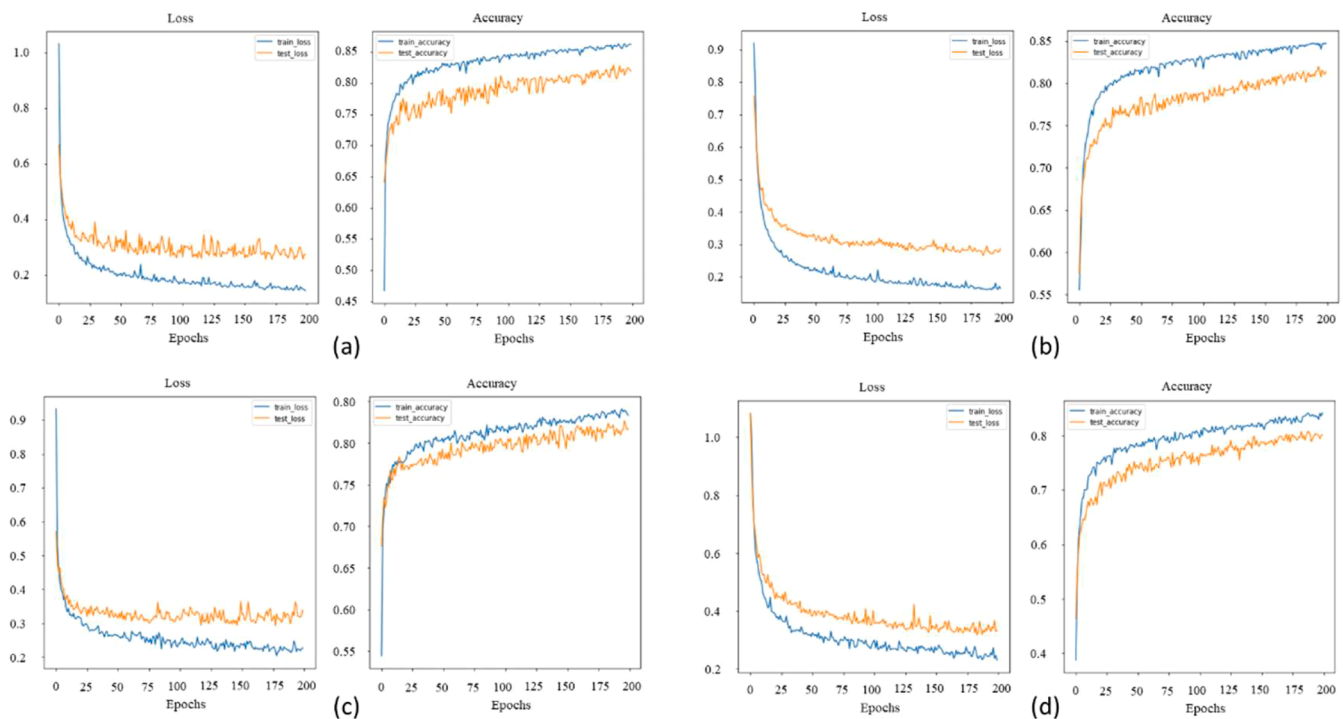


Fig. 7. Performance of four deep learning models without transfer learning. (a) Densenet-121, (b) Inception V3, (c) Resnet-50, (d) VGG19.

Table 7

Class-specific performance metrics of Resnet-50 without transfer learning.

Classes	Precision	Recall	F1-score
Angular Leafspot	0.848	0.822	0.835
Anthracoese Fruit	0.847	0.820	0.833
Gray Mold Fruit	0.871	0.822	0.846
Gray Mold Leaf	0.840	0.820	0.83
Healthy Fruit	0.798	0.822	0.81
Healthy Leaf	0.691	0.817	0.749
Powdery Mildew Fruit	0.761	0.823	0.791
Powdery Mildew Leaf	0.869	0.822	0.845

Type 1 and Type 2 configurations. The number of trainable parameters for Type 1 models was in the thousands, while the number of trainable parameters for Type 2 configuration was in the millions. Despite Type 3 configurations having an equal number of trainable parameters as Type 2, the larger size of the training dataset and trainable parameters led to increased training times in the case of Type 3. Detailed information regarding training time, size, and number of parameters for each model is tabulated in Table 7.

Despite having significant differences in depth, size, and number of trainable parameters, Densenet121 and Inception V3 demonstrated comparable performance in the Type 1 configuration. However, Densenet121 outperformed in both Type 2 and Type 3 configurations. Densenet utilizes a dense connectivity pattern, where each layer receives feature maps from all previous layers, allowing for efficient feature reuse and promoting gradient flow through the network. These dense connections mitigate the vanishing gradient problem and enable lower layers to directly access gradients during training. Consequently, Densenet effectively captures and propagates information across different depths. In the case of Inception, it utilizes a multi-scale approach to feature learning by employing various kernel sizes within its inception modules. This design enables the network to capture features at different scales and abstraction levels simultaneously. Inception modules use 1×1 , 3×3 , and 5×5 convolutions, effectively capturing fine-grained and coarse-grained features (Huang et al., 2022). The improved performance

of Densenet121, despite its smaller number of parameters and input size, can be attributed to its cross-layer connectivity, which concatenates the features from previous layers rather than adding them. Moreover, the Densenet promotes enhanced information flow throughout the network by directly incorporating gradients from all layers through the loss function, thereby mitigating overfitting. A study by Singh et al. (2021) reported Densenet201 achieving 79.01 % and InceptionV3 achieving 75.93 % accuracy in detecting disease and pest infections in coconut trees. Similarly, while detecting 26 diseases across 14 crop plants, Too et al. (2019) reported 99.76 % and 98.02 % accuracies for Densenet121 and InceptionV4, respectively.

In a comparison of model size, VGG19 was the largest model in size and number of trainable parameters, though it comprised only 19 layers. In the Type 1 configuration, VGG19 exhibited inferior performance compared to the other three models. However, it achieved similar accuracy to Inception V3 in the Type 2 configuration, though still less than Densenet's performance. Notably, when trained with augmented images, VGG19 outperformed Inceptionv3 and Densenet121, achieving 1.8 % and 0.9 % higher accuracy, respectively. The relatively lower performance of VGG19, despite its deep and large architecture with numerous parameters, may be attributed to its architectural limitations. Unlike Inceptionv3, which utilizes a multi-scale approach by using various kernel sizes to capture features at different scales and abstraction levels, and Densenet, which employs skip connections to facilitate gradient flow and mitigate vanishing gradient issues, VGG19 uses a simple architecture characterized by a repetitive pattern of small 3×3 convolutional filters and max-pooling layers. This simpler and homogeneous structure may lead to redundant feature learning, which learns similar or redundant representations of features across different layers.

This study showed the utility of transfer learning, improved model performance, and reduced training time, ultimately saving time and resources. However, it is essential to note that the performance of each model can vary depending on the specific task, and transfer learning excels when the new task shares similarities with the previous one. Considering the nature and requirements of disease detection in strawberries, transfer learning emerges as a suitable choice. While the current results demonstrate promising performances, further research should be

Table 8

Depth, size, trainable parameters, and training time for developed models.

Models	Depth	Size (MB)	No. of trainable parameters (million)			Training time (min)		
			Type 1	Type 2	Type 3	Type 1	Type 2	Type 3
Densenet-121	121	27.79	8.2×10^{-3}	6.962	6.962	66.82	116.2	1267.21
Inception V3	48	98.50	16.39×10^{-3}	25.128	25.128	85.53	149.62	1622.03
Resnet-50	50	92.20	16.39×10^{-3}	23.524	23.524	61.61	105.93	1168.401
VGG19	19	545.33	32.7×10^{-3}	139.603	139.603	83.65	161.69	1586.37

conducted to evaluate these models across different datasets to ensure their robustness and generalizability in various scenarios.

5. Conclusion

This research investigated the performance of four distinct deep learning architectures, namely Densenet-121, Inception V3, Resnet-50, and VGG-19, for disease detection in strawberry leaves and fruits. These architectures were adapted using transfer learning techniques encompassing feature extraction and fine-tuning and were trained with original images. Furthermore, the models were also trained without transfer learning using augmented images, and the performance of the models was assessed across three different configurations. The study outcomes indicated that Resnet-50 consistently achieved higher accuracy across all three scenarios, with its peak accuracy of 94.4 % attained through fine-tuning, while the least training time was observed when used as a feature extractor. Furthermore, the study demonstrated that the capability of each model to identify strawberry diseases was notably increased through fine-tuning compared to that of only utilizing them as feature extractors. Importantly, the adoption of transfer learning reduced training time significantly and resulted in superior performance compared to models trained from scratch. Each of the four models exhibited competitive performance in the classification task. The findings of this study can serve as a reference for practitioners and researchers, facilitating the selection of the appropriate model for strawberry disease detection based on specific application requirements and performance benchmarks. In the future, further studies could explore the effectiveness of these models in identifying diseases across the different crop species, emphasizing achieving higher classification performance. Additionally, subsequent works may investigate the practical deployment of developed models using hardware suitable for real field conditions, thus providing further validation for their utility (Table 8).

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CRediT authorship contribution statement

Sijan Karki: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Jayanta Kumar Basak:** Writing – review & editing, Validation. **Niraj Tamrakar:** Writing – review & editing, Validation. **Nibas Chandra Deb:** Writing – review & editing, Data curation. **Bhola Paudel:** Writing – review & editing, Formal analysis. **Jung Hoo Kook:** Resources. **Myeong Yong Kang:** Resources. **Dae Yeong Kang:** Project administration. **Hyeon Tae Kim:** Writing – review & editing, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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